

Power Quality Analyser – Logbook

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Design Rationale:

I used a waveform sample struct because every row in the csv file contained multiple numbers that belonged together in order so storing them together made it easier to later organise and pass the data into functions.

I split the code into the files I did because main.c was there to control the whole program and execute all the functions waveform.c was to define all the functions to do with calculations io.c was to define all the functions to do with file handling such as opening reading and writing to a file and then waveform.h for declaration of the calculation functions and finally io.h for declaration of the file handling functions.

I chose not to include any of the extra features as i wanted to meet the pass criteria properly and the extra features may have led to more complications and debugging.

Pointer and memory management:

I used malloc because I needed to allocate in this case some temporary memory for each row in the csv file so it could be loaded properly and efficiently so I wouldn't have the issue of using too much or too little memory which could have caused the program to crash or to use up an unnecessary amount of memory and become inefficient.

I used pointers in a few places in the program. The main pointer was the waveform sample pointer returned by malloc which pointed to the first sample in the array, I also used pointer parameters in functions so the CSV data could be passed into functions without copying the whole array, the file pointers such as FILE * were also used when opening, reading and writing files so that the program knows which file to use and the file pointer could then be passed into various functions like fgets, fprintf and fclose.

I used the free () calls to release the allocated malloc memory if I did not do this it would cause a memory leak and may then cause the program to crash.

Testing:

I tested the code with an empty CSV file, proper CSV file, a file that doesn't exist and no file/ too many files provided at all to make sure it recognises this and prints the correct error message. Also, whilst coding the project I had the row count and rows loaded printed in the console while testing which helped me make sure that the program was reading the same number of rows that it counted so I could debug it if there was a problem.

Empty CSV file:

```
Error: No CSV data rows found.  
  
Process finished with exit code 1
```

Correct CSV file:

```
Power Quality Waveform Analyser  
Input file: power_quality_log.csv  
Rows counted: 1000  
Rows loaded: 1000
```

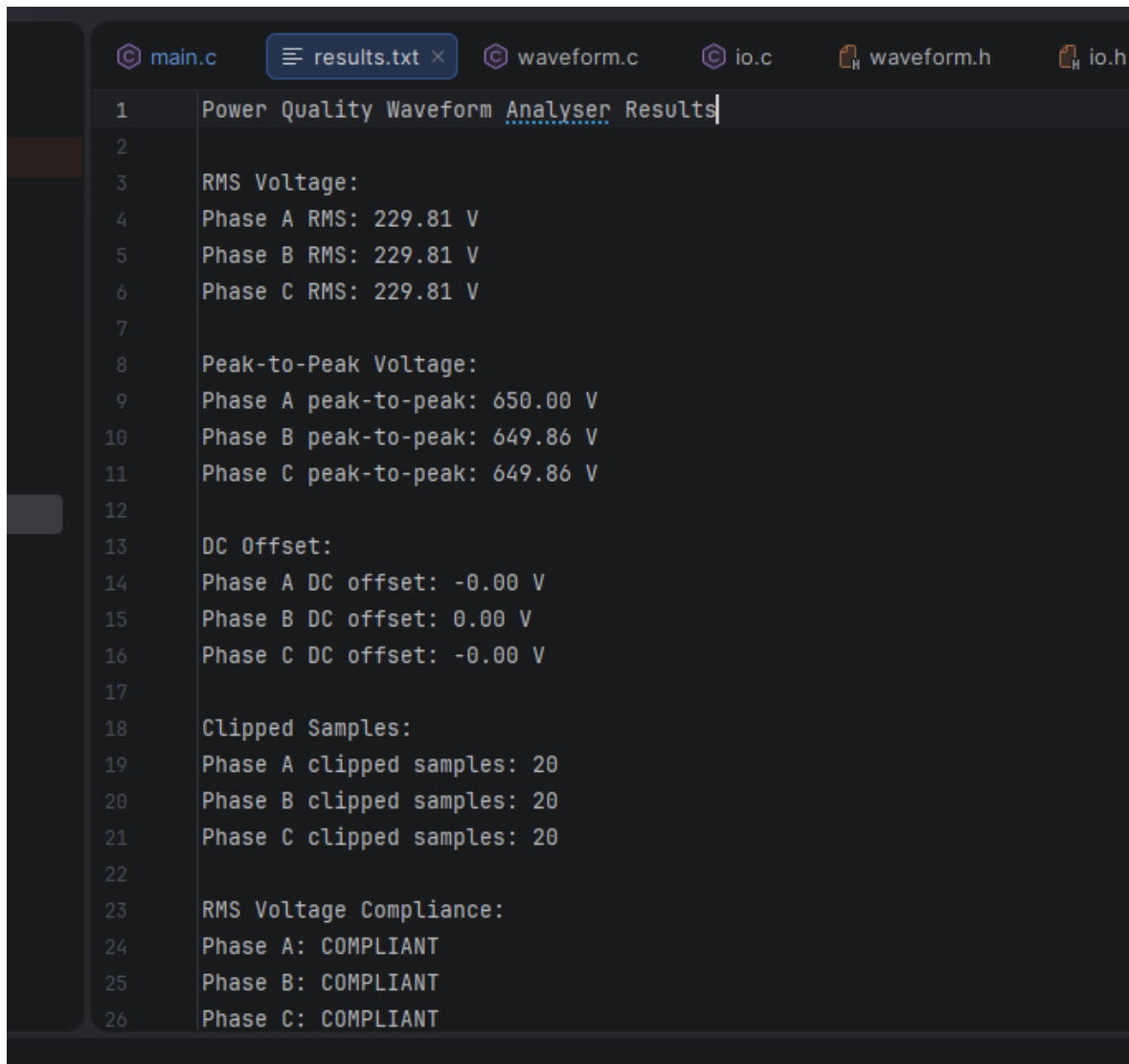
Non existing CSV file:

```
Error: File not found 'missingfile.csv'.  
  
Process finished with exit code 1
```

No file provided / too many files provided:

```
Usage: C:\Users\jvara\CLionProjects\PowerQualityAnalyser\cmake-build-debug\PowerQualityAnalyser.exe <csv_filename>  
Error: Please provide exactly one CSV filename.  
  
Process finished with exit code 1
```

Results File:



```
1 Power Quality Waveform Analyser Results
2
3 RMS Voltage:
4 Phase A RMS: 229.81 V
5 Phase B RMS: 229.81 V
6 Phase C RMS: 229.81 V
7
8 Peak-to-Peak Voltage:
9 Phase A peak-to-peak: 650.00 V
10 Phase B peak-to-peak: 649.86 V
11 Phase C peak-to-peak: 649.86 V
12
13 DC Offset:
14 Phase A DC offset: -0.00 V
15 Phase B DC offset: 0.00 V
16 Phase C DC offset: -0.00 V
17
18 Clipped Samples:
19 Phase A clipped samples: 20
20 Phase B clipped samples: 20
21 Phase C clipped samples: 20
22
23 RMS Voltage Compliance:
24 Phase A: COMPLIANT
25 Phase B: COMPLIANT
26 Phase C: COMPLIANT
```

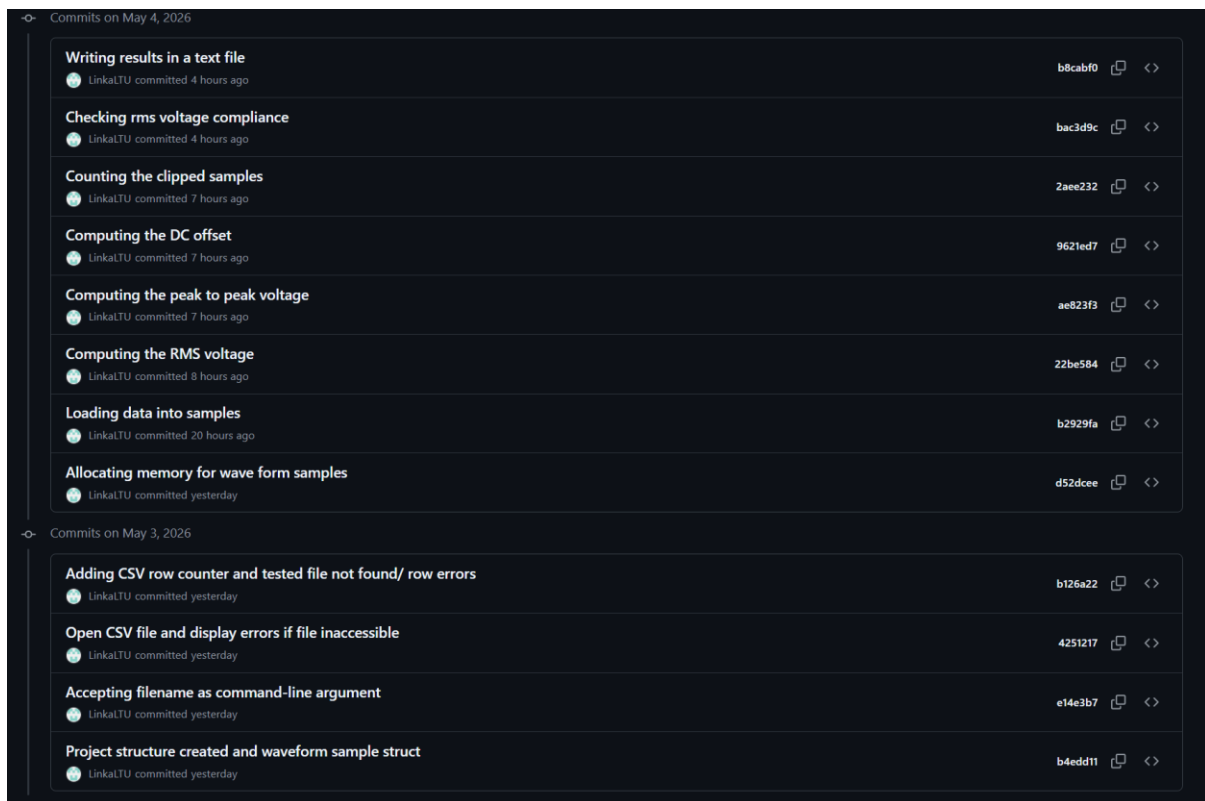
This showed that the program successfully wrote the calculated values to results.txt. The RMS values were 229.81 V for all three phases which is close to the expected 230 V RMS supply and inside the 207 V to 253 V compliance range and the peak-to-peak values were around 650 V which matches the expected waveform size and the DC offsets were close to 0 V which was expected as it is an AC waveform. The program also detected clipped samples and marked all phases as compliant as they were within the +-10%.

Debugging:

One bug I had was after renaming the project from untitled5 to PowerQualityAnalyser. CLion was still trying to run the old project name which caused the program to fail when I tried to run it. I realised this was because the CMake/project configuration had not updated properly. To fix it I updated CMakeLists.txt, reloaded the CMake project and changed the run configuration to PowerQualityAnalyser.

Another bug I had was with the condition in one of my file readings while loops. I accidentally had the loop checking `fgets(...) == NULL` which was the wrong way round because that means it is checking for when no line has been read. This made the file reading does not work properly. I fixed it by changing it to `fgets(...) != NULL` so the loop only runs while a line is being read from the CSV file.

Commits / Version control:



The screenshot shows a GitHub commit history interface. It is divided into two sections: 'Commits on May 4, 2026' and 'Commits on May 3, 2026'. Each section contains a list of commits with their titles, commit times, commit hashes, and icons for cloning and viewing the diff.

Commits on May 4, 2026			
Writing results in a text file	LinkaLTU committed 4 hours ago	b8cabf0	 
Checking rms voltage compliance	LinkaLTU committed 4 hours ago	bac3d9c	 
Counting the clipped samples	LinkaLTU committed 7 hours ago	2aee232	 
Computing the DC offset	LinkaLTU committed 7 hours ago	9621ed7	 
Computing the peak to peak voltage	LinkaLTU committed 7 hours ago	ae823f3	 
Computing the RMS voltage	LinkaLTU committed 8 hours ago	22be584	 
Loading data into samples	LinkaLTU committed 20 hours ago	b2929fa	 
Allocating memory for wave form samples	LinkaLTU committed yesterday	d52dcee	 
Commits on May 3, 2026			

Adding CSV row counter and tested file not found/ row errors	LinkaLTU committed yesterday	b126a22	 
Open CSV file and display errors if file inaccessible	LinkaLTU committed yesterday	4251217	 
Accepting filename as command-line argument	LinkaLTU committed yesterday	e14e3b7	 
Project structure created and waveform sample struct	LinkaLTU committed yesterday	b4edd11	 

I used GitHub for version control throughout the project making commits every time I added a working bit of code to the program which helped me keep track of the changes I made when I was working on the next commit showing what has been edited or added and it shows the development of the project over time.

References:

University of the West of England. (2025-26). UGMFGT-15-1 Programming for Engineers Coursework Brief: Power Quality Waveform Analyser.

University of the West of England. (2025-26). AC Waveforms and 3-Phase Power Reference Material.

Git. Git Documentation. Available at: <https://git-scm.com/doc>

cppreference. C reference. Available at: <https://en.cppreference.com/w/c>